

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2012

CL307 Structural Analysis-III

Date: 08.05.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

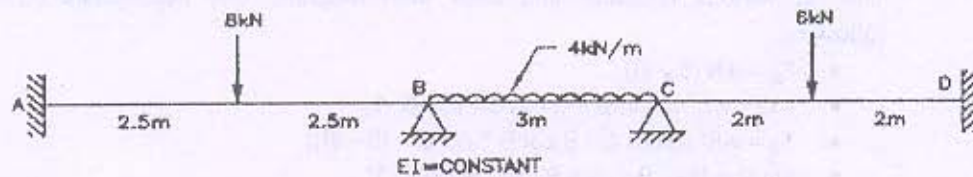
Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 A prestressed concrete beam, 200 mm wide and 300 mm deep, is used over an effective span of 6 m to support an imposed load of 4 kN/m. The density of concrete is 24 kN/m^3 . At the centre of span section of the beam, find the magnitude of the concentric prestressing force necessary for 100 mm zero fiber stress at the soffit when the beam is fully loaded. [07]

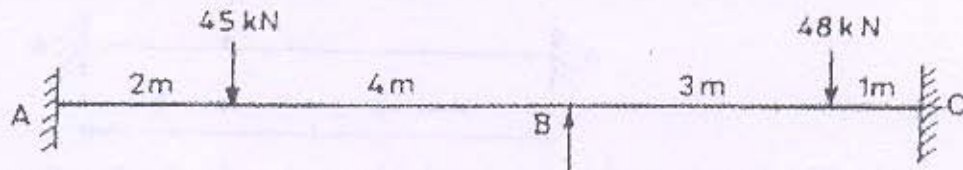
Q - 2 (a) Enlist advantages of prestressed concrete. [02]
(b) For a continuous beam ABCD shown in figure below, draw B.M. diagram. Support A rotates by $1/250$ radians in clockwise direction. Support B sinks by 30 mm. Support C sinks by 20 mm. $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 38.20 \times 10^5 \text{ mm}^4$. (Use Slope Deflection Method) [12]



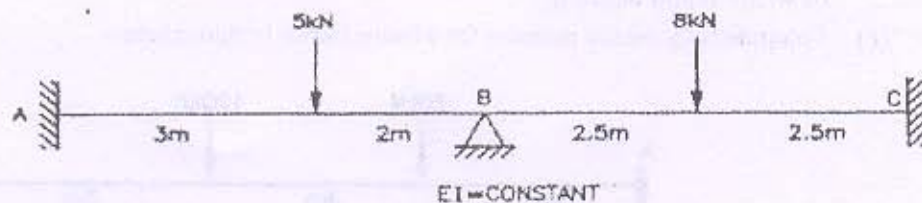
OR

(b) Determine the support moments and reactions for the continuous beams shown in below figures. Draw also S.F. and B.M. Diagram. (Use Slope Deflection Method)

(i) [06]

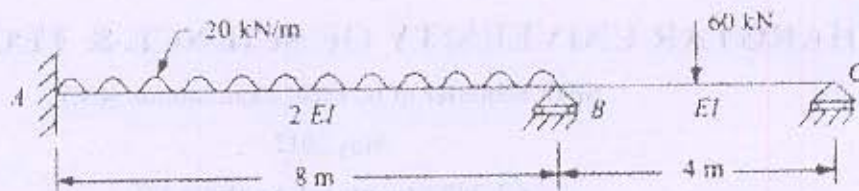


(ii) [06]



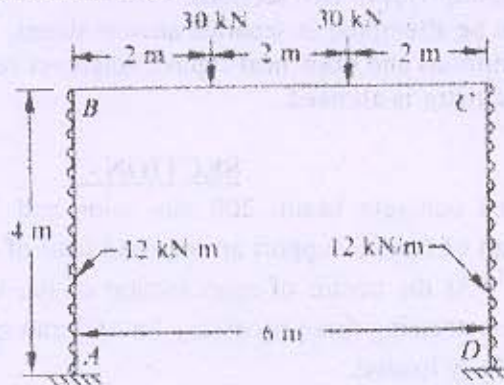
Q - 3 (a) Describe methods of prestressing. [02]

(b) Analyse the continuous beam shown in figure below by moment distribution method, if support B sinks by 10 mm. Take $EI = 4000 \text{ kNm}^2$. Draw bending moment diagram. [12]



OR

- (b) Analyse the frame shown in figure below by moment distribution method. Take EI as constant. Draw Bending moment diagram. [12]

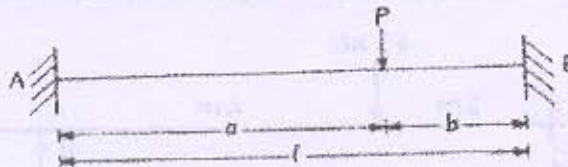


SECTION - II

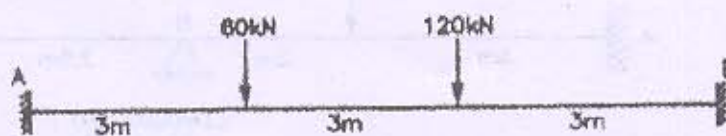
- Q - 4 A curved beam circular in plan symmetrically supported on six columns has radius 5m. The beam carries an udl of 30 kN/m including self-weight of beam. Determine S.F., B.M. and T.M. at various locations and draw their diagram. Use appropriate equation from the following: [07]

- $F_{\theta} = wR (\theta - \phi)$
- $M_{\theta} = wR^2 [\theta \sin \phi + \theta \cot \theta * \cos \phi - 1]$
- $T_{\theta} = wR^2 [\theta \cos \phi - \theta \cot \theta * \sin \phi - (\theta - \phi)]$
- $\sin \phi = \frac{[\sin^2 \theta + \cos \theta (\theta^2 - \sin^2 \theta)^{0.5}]}{\theta}$

- Q - 5 (a) Find fixed end moments for the fixed beam shown in figure by column analogy method. [05]



- (b) Find shape factor for a beam of circular section of radius R or for a rectangular beam of width b and depth d. [03]
- (c) Compute fully plastic moment for a beam shown in figure below. [06]



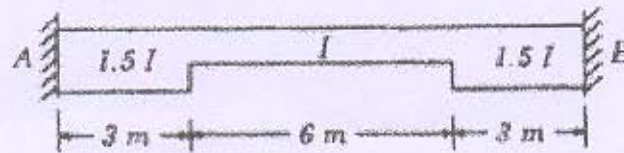
OR

- (c) A fixed beam of span 6m span carries a uniformly distributed load of 175 kN/m over left half of the span. Determine the fully plastic moment for the beam. Also calculate [06]

plastic section modulus required. $F_y = 250 \text{ MPa}$.

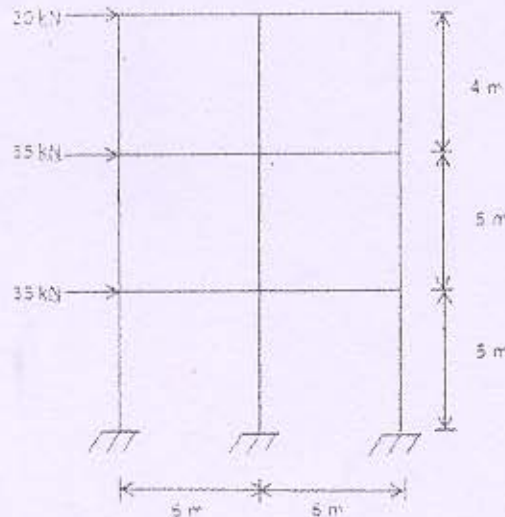
Q - 6 (a) Find the stiffness and carry-over factor for the fixed beam shown in figure below.

[05]



(b) Analyse the portal frame shown in figure below by portal method.

[09]



OR

(b) Analyse the frame shown in figure by cantilever method.

[09]

